

DFA – User Authentication Workflow

Transition Table

State	Input	Next	Meaning
q_0	Register/Login	q_1	Open authentication
q_1	Credentials	q_2	Credentials entered
q_2	Correct	q_3	Credentials valid
q_2	Wrong	q_5	Credentials invalid
q_5	Retry	q_1	Try again
q_3	Access	q_4	Authentication successful
q_4	Logout	q_6	Session terminated
q_6	End	q_7	End state

Why this is a DFA

For each current state and input, exactly one next state is selected. This makes the transition deterministic.

